

Ontological Fluency in Physics

Diagnostic and Classroom Tasks for Reading Equations, Models and Explanations

Physics problem solving requires more than remembering equations and rearranging symbols. Students must also recognise what kind of thing each symbol represents, how the quantities belong to a system, and what operations the model permits.

STUDENT WORKBOOK

Name: _____

Class: _____

Teacher: _____

Date started: _____

How to use this workbook

This workbook treats recurring physics errors as more than failures of memory. A student may know a formula and still misunderstand what the symbols denote. The tasks therefore ask you to classify quantities, repair misleading sentences, expand equations into relations, diagnose wrong solutions and distinguish observation from model.

The central sequence used throughout is:

phenomenon → system → entities → relations → quantities → equation → operation

At each stage, ask:

1. What is happening?
2. What system is being analysed?
3. What objects or structures matter?
4. What interacts with what?
5. Which quantities represent those relations?
6. Which equation is justified?
7. What mathematical operation follows?

Diagnostic prompt

When an answer is wrong, do not ask only, “Which fact was forgotten?” Also ask:

What must the solver believe this quantity *is* for the answer to make sense?

A model exemplar before the physics: the crowded corridor

Model exemplar

Imagine a school corridor during lesson changeover. Fifty students occupy a corridor section of area 40 m^2 . A second corridor section contains eighty students in 100 m^2 . We may define a simple crowd-density model:

$$D = \frac{N}{A},$$

where N is the number of people in the selected region and A is the floor area of that same region. For the first corridor,

$$D_1 = \frac{50}{40} = 1.25 \text{ people m}^{-2}.$$

For the second,

$$D_2 = \frac{80}{100} = 0.80 \text{ people m}^{-2}.$$

The second corridor contains more people, yet it is less crowded. The relevant property is the ratio, not the amount alone.

This mundane example already contains most of the reasoning demanded later in physics.

1. **Phenomenon:** people occupy a corridor.

2. **System:** one clearly bounded corridor region.
3. **Entities:** people and floor area.
4. **Relation:** number of people per unit area.
5. **Quantities:** N , A , and D .
6. **Equation:** $D = N/A$.
7. **Operation:** divide the number belonging to the region by the area of that same region.

Guided prompts

MYP route: Complete the sentence: “The second corridor has more people, but it is less crowded because ...”

DP extension: Explain why crowd density is an intensive ratio property within this simplified model. State one respect in which the model fails to capture the lived experience of congestion.

What the exemplar teaches

- A quantity can be a **ratio property**, not an amount.
- The numerator and denominator must refer to the **same system**.
- A familiar verbal description, such as “how crowded it is”, does not by itself guarantee operational fluency.
- The equation is a **model**: it ignores walking direction, speed, clustering, bags, doorways and personal space.
- A symbol becomes useful only when the student knows what kind of quantity it represents and how it is measured.

The corridor model is deliberately tangential. It gives students a familiar case in which the ontology is easier to see before the same reasoning is required for density, current, power, force and field.

1. What kind of thing is this?

Concept

Ontological fluency is the ability to recognise whether a physics term is functioning as an object, property, interaction, state, process, rate, ratio, field, model parameter or representational construct.

1.1 A working classification

Complete the final column.

Category	Guiding question	Your example
Object	What physical system or body is selected?	
Property	What characteristic is assigned to a system?	
Interaction	What acts between systems?	
State	What describes the system at an instant?	
Process	What is changing or being transferred?	
Rate	What changes per unit time?	
Ratio property	What quantity is defined as one amount per another?	
Field	What quantity is assigned to locations?	
Model parameter	What number characterises a model or component?	
Representation	What symbol, arrow, line or diagram stands for something else?	

1.2 Classify the terms

Guided prompts

MYP route: Use the sentence stem: “___ is best classified as a ___ because it tells us ___.”

DP extension: Where two classifications are plausible, explain how the term’s role changes with the model or context.

Place each term into the category that best describes how it functions in the stated context. Some terms may fit more than one category; justify your choice.

mass force acceleration density current

power energy electric field resistance field line

Task

For each term, write one classification and one reason.

Common trap

The grammar of a noun does not guarantee that the word denotes a thing. “Acceleration”, “current” and “power” are nouns, yet each denotes a rate or change.

2. The noun beneath the equation

2.1 Density: a simple equation with hidden commitments

Bridge from the corridor model

In the corridor example, $D = N/A$ described people per unit area. Density uses the same ratio structure:

$$\rho = \frac{m}{V}.$$

Read this first as “mass per unit volume”, then ask what physical system supplies both measurements.

Density is often described as “the amount of stuff packed into a given volume”. The phrase is memorable, but it leaves several questions unresolved.

Task

Answer each question.

1. What physical system must m and V refer to?
2. Is ρ an amount, a rate or a ratio property?
3. Why can a large block and a small block have the same density?
4. Why is ρ not simply another word for mass?
5. What does the dimensional relation

$$[\rho] = ML^{-3}$$

tell you?

Diagnostic prompt

A student says:

A block has twice the mass, so it must have twice the density.

What must the student be assuming about density?

Task: Same material, different sample

Two aluminium blocks have volumes 20 cm^3 and 50 cm^3 . The smaller block has mass 54 g.

1. Calculate the density of aluminium.
2. Predict the mass of the larger block.
3. Explain why density remains unchanged while mass and volume change.

2.2 Newton's second law: what does the equation suppress?

Model exemplar: two people pushing a trolley

A trolley is pushed to the right by one person with 60 N and to the left by another with 20 N. The trolley's mass is 10 kg.

The physically relevant force is not either push alone:

$$F_{\text{net}} = 60 - 20 = 40 \text{ N}.$$

Hence

$$a = \frac{F_{\text{net}}}{m} = \frac{40}{10} = 4 \text{ m s}^{-2}.$$

The example makes visible what $F = ma$ suppresses: several interactions, direction, vector addition and the selection of one system.

Compare:

$$F = ma,$$

$$F_{\text{net}} = ma,$$

$$\sum \mathbf{F} = m\mathbf{a},$$

$$\sum_i \mathbf{F}_i = \frac{d\mathbf{p}}{dt}.$$

Task

Complete the table.

Expression	What it makes explicit	What the reader must still supply
$F = ma$		
$F_{\text{net}} = ma$		
$\Sigma \mathbf{F} = m\mathbf{a}$		
$\Sigma_i \mathbf{F}_i = d\mathbf{p}/dt$		

Diagnostic prompt

Which expression is easiest to use? Which is most physically explicit? Why are these not necessarily the same?

Task: Repair the possession grammar

Rewrite each sentence so that the interaction is explicit.

1. The object has a force of 20 N.
2. The book has weight.
3. The car has a forward force.
4. The wire has current.

3. Five families of ontological error

3.1 Substance errors

Model exemplar: people passing through a doorway

Suppose 30 people pass through a doorway in 10 s. The flow rate is

$$R = \frac{30}{10} = 3 \text{ people s}^{-1}.$$

The rate is not a substance stored in the doorway. It describes how much crosses a boundary per unit time. Current and power require the same kind of reading.

A substance error occurs when a process, rate or bookkeeping quantity is treated as material stuff.

Task

For each statement:

1. underline the word that suggests a substance;
 2. state what the quantity is more precisely;
 3. write an equation that clarifies the distinction.
1. Current is used up by the bulb.
 2. Heat is stored in the metal.
 3. Electricity flows out of the battery.
 4. Energy is contained inside the appliance.

Concept

Useful equations include

$$I = \frac{\Delta Q}{\Delta t}, \quad P = \frac{\Delta E}{\Delta t}.$$

A rate may describe a flow or transfer without itself being a material substance.

Model exemplar: the shortcut that “wants” to be used

Likewise, “heat seeks equilibrium” or “light chooses the quickest path” can be useful shorthand, but the grammatical agent must not be mistaken for a mechanism.

Task

1. Gravity pulls the object down.
2. Inertia keeps the object moving.
3. The battery pushes electrons around the circuit.
4. Heat seeks equilibrium.
5. Light chooses the path of least time.

[illegible]

3.3 Attribute errors

An attribute error occurs when a relation is treated as a possession.

Compare:

object $\rightarrow$ has $\rightarrow$ force

with

Earth $\rightarrow$ exerts gravitational force on $\rightarrow$ object.

Task

Convert each weak triple into a more informative subject–predicate–object graph.

1. object $\rightarrow$ has $\rightarrow$ weight
2. wire $\rightarrow$ has $\rightarrow$ current
3. charge $\rightarrow$ has $\rightarrow$ electric field
4. magnet $\rightarrow$ has $\rightarrow$ repulsion

3.4 Process–state errors

Model exemplar: distance and speed

A journey length of 120 m is an amount of distance. A speed of 6 m s^{-1} is a rate. The unit itself records the ontology: metres per second.

The same reading applies to amperes as coulombs per second and watts as joules per second.

Classify each quantity as an amount, state, rate, change or transfer.

charge	current	potential difference
energy	power	heat transfer
velocity	acceleration	displacement
temperature	internal energy	density

Task

Use the units as clues.

$$A = \text{C s}^{-1}, \quad W = \text{J s}^{-1}, \quad \text{m s}^{-2} = \frac{\text{m s}^{-1}}{\text{s}}.$$

What does each unit reveal about the ontology of the quantity?

3.5 Model–object errors

Model exemplar: a route on a map

A line drawn on a map is not a physical road. It represents the road while suppressing width, surface texture, gradients, traffic and temporary closures.

Field lines, rays and force arrows play a comparable representational role. They are indispensable for reasoning, but they are not the physical phenomena themselves.

Task

For each statement, identify the representation and the phenomenon.

1. Magnetic field lines leave the north pole and enter the south pole.
2. A ray shows the path travelled by a photon.
3. A force arrow is attached to the object.
4. A real gas disobeys the ideal gas law.

4. Equation unpacking

Fully worked exemplar: a kettle

A 2.0 kW kettle transfers energy for 60 s.

1. **Phenomenon:** electrical energy is transferred to the kettle and its contents.
2. **System:** the kettle and water, unless another boundary is specified.
3. **Entities:** supply, heating element, water and surroundings.
4. **Relations:** the supply transfers energy; some reaches the water and some the surroundings.
5. **Quantities:** power P , energy E , time t .
6. **Equation:** $P = E/t$, hence $E = Pt$.
7. **Operation:**

$$E = (2000 \text{ J s}^{-1})(60 \text{ s}) = 1.2 \times 10^5 \text{ J.}$$

The calculation is short because the modelling decisions have been made explicit first.

Use the same sequence for every problem:

phenomenon $\rightarrow$ system $\rightarrow$ entities $\rightarrow$ relations $\rightarrow$ quantities $\rightarrow$ equation $\rightarrow$ operation.

4.1 Worked scaffold: a falling object

A ball is released near Earth's surface. Air resistance is neglected.

Task

Complete the scaffold.

Phenomenon

System

Entities

Relations

Quantities

Equation

Operation

Draw or write an SPO graph linking Earth, the ball, gravitational force and acceleration.

A block is pulled horizontally at constant velocity.

Complete the full seven-stage sequence.

[illegible]

5. Diagnosing another student's reasoning

How to diagnose a wrong answer

Use this four-step routine:

1. Quote the claim that is physically wrong or incomplete.
2. Infer what kind of thing the student thinks the quantity is.
3. Replace that ontology with a more useful one.
4. Rebuild the explanation using an equation, diagram or unit.

Mini-example: “The bulb uses up current.”

The student treats current as consumable stuff. Reclassify current as charge transferred per unit time, then distinguish it from power, the rate of energy transfer.

5.1 Motion requires a forward force

A student writes:

The car is moving forwards, so there must be a forward resultant force.

Task

1. What has the student confused?
2. What ontology of force would make this answer seem reasonable?
3. What would be true if the car moved at constant velocity?
4. Write the relevant vector equation.

5.2 Heavier objects fall faster

A student writes:

The heavier object has greater weight, so it accelerates faster.

Task

Use

$$F_g = mg \quad \text{and} \quad a = \frac{F_{\text{net}}}{m}$$

to diagnose and repair the reasoning.

5.3 Current is consumed

A student writes:

The current after a bulb is smaller because some electricity has been converted into light.

Task

1. Which quantity is being treated as a substance?
2. Which quantity is actually transferred to the surroundings?
3. Which equation distinguishes current from power?
4. Rewrite the explanation.

5.4 Power is an amount

A student correctly calculates

$$P = \frac{600 \text{ J}}{30 \text{ s}} = 20 \text{ W}$$

and then says:

The appliance contains 20 W of energy.

Task

Explain why the arithmetic is correct but the ontology is wrong.

5.5 Density belongs to size

A student writes:

The larger cube is denser because it contains more matter.

Task

Explain the confusion between extensive and intensive quantities.

5.6 A force arrow is a thing

A student says:

The arrow on the free-body diagram is the force itself.

Task

Distinguish:

1. the interaction;
2. the physical quantity;
3. the vector representation.

6. From description to explanation

Model exemplar: three levels of answer

Question: Why does a metal spoon become warm in hot soup?

- **Description:** the spoon's temperature rises.
- **Predictive relation:** energy transfer depends on temperature difference, material and geometry.
- **Mechanistic account:** microscopic interactions redistribute energy through the material.

A stronger answer does not merely add words. It connects the observation to a model at an appropriate level.

6.1 Case A: Muons reaching Earth's surface

Cosmic rays strike atoms high in the atmosphere and produce pions. Pions decay into muons. Muons have a proper lifetime of about $2.2 \mu\text{s}$, yet many reach Earth's surface.

From Earth's frame, time dilation increases the measured lifetime. From the muon's frame, length contraction reduces the atmospheric distance. Both descriptions are required by Lorentz symmetry.

Task

Classify each statement as observation, measurement, inference, model or explanation.

1. Muon tracks are detected near Earth's surface.
2. The proper lifetime is approximately $2.2 \mu\text{s}$.
3. The atmosphere is length-contracted in the muon's frame.
4. Lorentz symmetry makes the two frame descriptions consistent.

Diagnostic prompt

Why does this case offer relatively strong mechanistic closure?

6.2 Case B: Why is ice slippery?

Candidate mechanisms include:

- a quasi-liquid surface layer;
- pressure melting;
- frictional heating;
- molecular disorder and mobility at the interface.

No single account explains every temperature, pressure and surface condition.

Task

1. Why is “slipperiness” not a simple intrinsic property?
2. Which conditions determine which mechanism matters?
3. Why might a patchwork explanation be scientifically stronger than one universal story?

6.3 Case C: Magnetic repulsion

When like poles are pushed together, a mechanical resistance is felt. A classical account describes the change in field energy and the force associated with its gradient. A quantum account invokes electromagnetic interaction at particle level.

$$\mathbf{F} = \nabla(\mathbf{m} \cdot \mathbf{B})$$

Task

Fill each blank with one word from the bank.

The mechanical _____ is interpreted as a force. Virtual photons _____ electromagnetic interaction. What we _____ as repulsion is a macroscopic result. The field model is effective within a _____.

Word bank: experience, domain, sensation, mediate

Task

What does the model explain successfully? Where does the intuitive demand for a literal “pushing agent” exceed what the model provides?

6.4 Comparative explanation table

Guided prompts

MYP route: For each case, complete: “We observe . . . ; the model explains this by . . . ; it does not yet explain . . .”

DP extension: Distinguish explanatory unification, domain-limited mechanism and foundational stopping point.

Task			
Complete the final three columns.			
Case	Observation	Explanatory structure	Limit
Muon survival			
Slippery ice			
Magnetic repulsion			

7. Knowledge graphs and ontology

Model exemplar: a classroom relation

Compare:

student $\rightarrow$ has $\rightarrow$ feedback

with

teacher $\rightarrow$ gives feedback to $\rightarrow$ student.

The second triple reveals an interaction and its direction. In physics, replacing “has” with a more precise predicate often exposes the missing mechanism or relation.

Knowledge graphs represent statements as subject–predicate–object triples. Their structure can expose assumptions hidden by ordinary sentences.

7.1 Classify the edge

Mark each triple as primarily:

measured inferred defined modelled assumed.

Task

1. electron $\rightarrow$ has charge $\rightarrow -e$
2. photon $\rightarrow$ is modelled as $\rightarrow$ field excitation
3. proton $\rightarrow$ consists of $\rightarrow$ quarks and gluons
4. magnetic field $\rightarrow$ has lines $\rightarrow$ loops
5. Sirius $\rightarrow$ is located at $\rightarrow$ measured coordinates

7.2 Repair the predicate

Task

Replace the weak predicate “has” with something more informative.

1. electron → has → electric field
2. magnet → has → repulsion
3. wire → has → current
4. object → has → acceleration

7.3 Realist and model-based readings

Task

Choose one concept: force, field, mass, charge, photon or spin.

Write:

1. a realist interpretation;
2. a model-based interpretation;
3. a use-based interpretation focused on how the term functions in physics practice.

8. Observation and reconstruction: the case of Sirius

We say that we see Sirius. More directly, photons enter an eye or detector from a narrow angular region. Coordinates, spectra, parallax, proper motion and repeated observations are integrated into a stable model of a star.

Task

Sort each item into one of four categories:

observation measurement inference name or convention.

1. photons detected at a telescope;
2. right ascension and declination;
3. atmospheric scintillation;
4. the continuing object called Sirius;
5. the statement that the photons came from one star;
6. the name "Sirius".

Diagnostic prompt

Does saying that Sirius is inferred mean that Sirius is unreal? Explain the difference between epistemic access and ontological commitment.

Task

Construct a knowledge graph showing how observations support the model of Sirius.

Worked exemplar: $P = E/t$

- **Words:** power is energy transferred per unit time.
- **Ontology:** E is an amount transferred; t is an interval; P is a rate.
- **Units:** $\text{W} = \text{J s}^{-1}$.
- **SPO form:**
system $\rightarrow$ transfers energy at $\rightarrow$ power.
- **Condition:** the average relation applies over the chosen interval; instantaneous power may vary.
- **Common error:** treating watts as an amount of energy.

$$V = IR, \quad E_k = \frac{1}{2}mv^2, \quad p = mv, \quad P = \frac{E}{t}, \quad Q = It.$$

1. Write the equation in words.
2. Identify the system.
3. Classify each symbol ontologically.
4. State the units of each quantity.
5. Draw or write an SPO graph.
6. Give one physical situation in which the equation applies.
7. Give one common misconception.
8. State one limitation or condition of use.

[illegible]

10. Final synthesis

Diagnostic prompt

A formula sheet removes the need to remember an equation. What important acts of interpretation remain?

Task

Complete the sentence:

A student has ontological fluency when they can . . .

Task

Return to one error you made earlier in the workbook.

1. What was your original interpretation?
2. What kind of thing did you think the quantity was?
3. What classification is now more useful?
4. How does the new classification change the equation or solution?

Self-assessment

Tick one box for each statement.

Statement	Not yet	Developing	Secure
I can distinguish an amount from a rate.	☐	☐	☐
I can identify when a relation is being treated as a possession.	☐	☐	☐
I can distinguish a model representation from a physical object.	☐	☐	☐
I can use units as clues to the meaning of a quantity.	☐	☐	☐
I can identify the system before choosing an equation.	☐	☐	☐
I can diagnose the ontology behind a wrong answer.	☐	☐	☐
I can move between words, diagrams, graphs and equations.	☐	☐	☐
I can state the limits of an equation or model.	☐	☐	☐

Final reflection

Which idea from this workbook most changed how you read physics equations?

Teacher answer key and misconception map

Teacher note

The answers below are indicative rather than exhaustive. The purpose of many tasks is to reveal the student's category system and language, not to enforce one final metaphysical taxonomy.

Core classifications

Mass	Usually treated as a system property and inertial parameter.
Force	Interaction represented by a vector; not a stored property of motion.
Acceleration	Rate of change of velocity relative to a frame.
Density	Ratio property m/V ; typically intensive for a homogeneous material.
Current	Rate of charge transfer, $I = \Delta Q / \Delta t$.
Power	Rate of energy transfer, $P = \Delta E / \Delta t$.
Field	Quantity assigned to positions in space and time.
Resistance	Relational/model parameter connecting potential difference and current.
Field line	Representational construct showing local field direction and relative strength.

Density diagnostic

A larger mass does not imply a larger density because density depends on the ratio m/V . For samples of the same homogeneous material, mass and volume scale together. The statement “amount of stuff packed into a volume” is useful but vague: it can lead students to confuse mass with density or treat density as a material substance.

Newton's second law

$$F = ma$$

suppresses net force, vector character, system choice and frame conditions.

$$F_{\text{net}} = ma$$

makes the resultant explicit.

$$\sum \mathbf{F} = m\mathbf{a}$$

makes vector addition explicit.

$$\sum_i \mathbf{F}_i = \frac{d\mathbf{p}}{dt}$$

shows the more general momentum relation.

Five error families

Substance	Rate, process or bookkeeping quantity treated as material stuff.
Agent	Abstract term granted causal agency or intention.
Attribute	Relation treated as a possession of one object.
Process–state	Rate, transfer or change confused with an amount or state.
Model–object	Diagram, line, ray, arrow or idealisation treated as the phenomenon itself.

Typical repairs

Current used up: Current is charge transferred per unit time. In a steady series circuit the rate is the same around the circuit; energy is transferred in components.

Inertia keeps motion going: With zero resultant force, acceleration is zero and velocity remains constant. Inertia is not an extra forward force.

Heavier objects fall faster: Although $F_g = mg$ is larger for greater m , the inertial response also contains m , giving $a = F_g/m = g$ when air resistance is neglected.

Power as energy: Watts are joules per second. Power is a rate, not an amount stored in an appliance.

Field lines: Field lines are graphical devices. The field is represented by a vector assigned to each point.

Explanation cases

Muon: strong closure because observations, measured lifetime and Lorentz transformations form a coherent, testable framework.

Slippery ice: patchwork explanation because several mechanisms matter under different temperature, pressure, speed and surface conditions.

Magnetic repulsion: field and quantum models predict and organise the interaction, while a demand for a familiar mechanical agent may remain unsatisfied.

Suggested assessment criterion

A secure response should normally include:

1. correct classification of the quantity;
2. explicit system identity;
3. relational rather than possession grammar where appropriate;
4. consistent units and dimensions;
5. distinction between model and phenomenon;
6. recognition of assumptions or domain.

Selected references

References

- [1] M. T. H. Chi, J. D. Slotta, and N. de Leeuw, “From things to processes: A theory of conceptual change for learning science concepts,” *Learning and Instruction*, vol. 4, no. 1, pp. 27–43, 1994.
- [2] M. Reiner, J. D. Slotta, M. T. H. Chi, and L. B. Resnick, “Naive physics reasoning: A commitment to substance-based conceptions,” *Cognition and Instruction*, vol. 18, no. 1, pp. 1–34, 2000.
- [3] J. Clement, “Algebra word problem solutions: Thought processes underlying a common misconception,” *Journal for Research in Mathematics Education*, vol. 13, no. 1, pp. 16–30, 1982.
- [4] B. L. Sherin, “How students understand physics equations,” *Cognition and Instruction*, vol. 19, no. 4, pp. 479–541, 2001.
- [5] E. Kuo, M. M. Hull, A. Gupta, and A. Elby, “How students blend conceptual and formal mathematical reasoning in solving physics problems,” *Science Education*, vol. 97, no. 1, pp. 32–57, 2013.
- [6] L. Wittgenstein, *Philosophical Investigations*. Oxford: Blackwell, 1953.
- [7] N. Cartwright, *How the Laws of Physics Lie*. Oxford: Clarendon Press, 1983.